Bellman Residual Minimization – How to do it right?

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Learning Theory Workshop, POSTECH

Intro



- Background
  - Bachelor in Mathematics (2011–2015), KAIST
  - Master & Ph.D. in Mathematics (2015–2022), KAIST

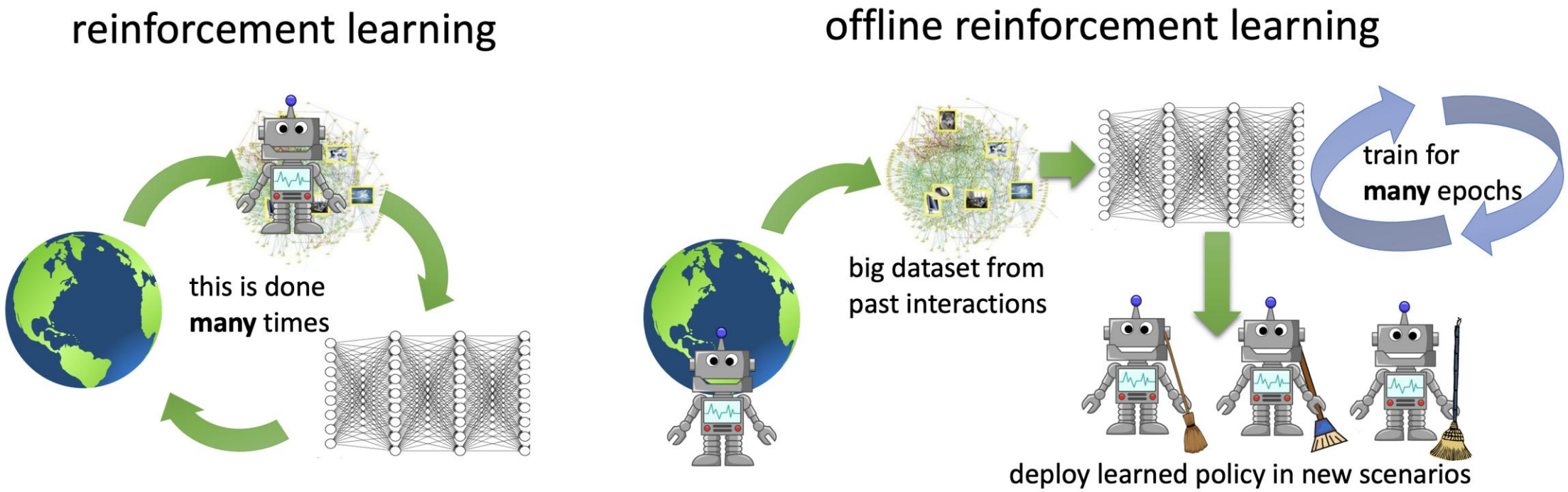
- Multiple postdocs
  - at Univ. of Arizona, with Prof. Kwang-Sung Jun (Mar 2022 ~ Aug 2023)
  - at NYU, with Prof. Junpei Komiyama (Aug 2023 ~ Feb 2024)
  - at Univ. of Milan, with Prof. Nicolò Cesa-Bianchi (Mar 2024 ~ Feb 2025)

Table of Contents

- Offline Reinforcement Learning
  - FQI, BRM, Others…
- BRM and Dual Approach
  - GLADIUS algorithm
  - Statistical error
- Trajectory BRM
  - Stoc-TBRM

Offline RL

Characteristics of Offline RL



Applications:

- Medical trials (when one trial is dangerous and expensive)
- Online Transactions (when you already have a big dataset)

Fundamental Equation

$$\boldsymbol{Q} = \mathcal{T}\boldsymbol{Q}$$

- Called **Bellman (Optimality) Equation**
  - Actually, most online RL Algorithms also rely on it.

- But it can be interpreted in various ways.

Fitted Q-Iteration

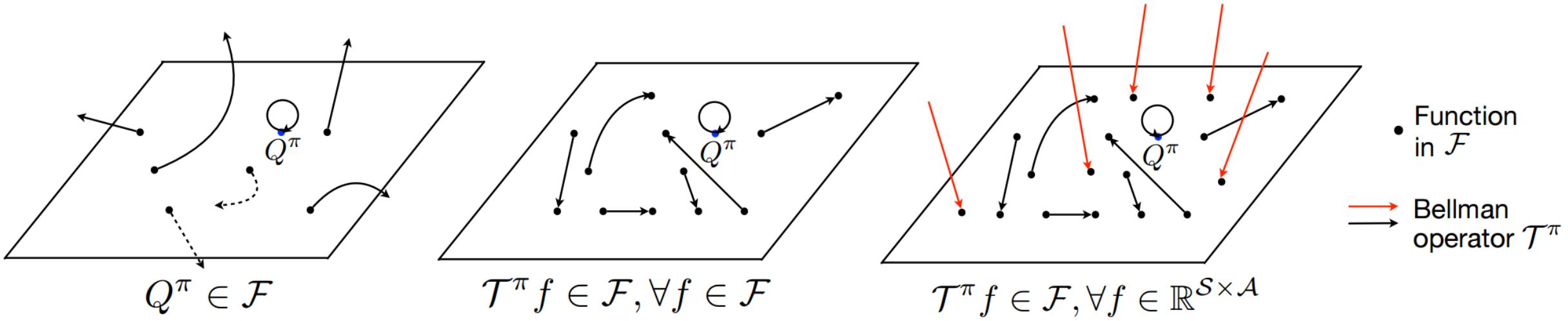
$$Q_{k+1} \leftarrow argmin_Q \sum_{(s,a,r,s') \in \mathcal{D}} \left(Q(s,a) - r - \gamma V_k(s')\right)^2$$

- Iterative method
- Straightforward, but can diverge even in a very strong condition!
  - Bellman Completeness is almost a necessary condition to succeed.

Proposition 1 (**Tsitsiklis and Van Roy, 1996**) *FQE can diverge even when all of the following hold:*

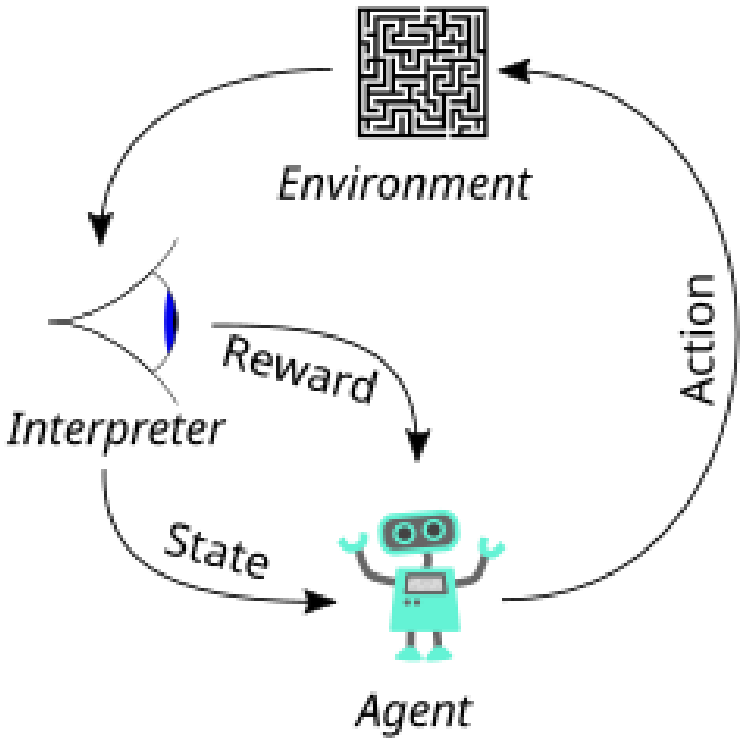
1. $|\mathcal{D}| = \infty$ *and minimization in Eq.(7) is exact.*
2. $\mathcal{F}$ *is a 1-dim linear function class that exactly captures* Q^π, *i.e.,* $Q^\pi \in \mathcal{F}$ *(a.k.a.* realizability*).*

Fitted Q-Iteration



Offline RL as Statistical RL

Textbook RL Diagram



But in Offline RL:

- No interaction
- No new data

… More like traditional ML (Statistical Learning)!

Importance Sampling (IS)

- Okay, we said offline RL is like stat ML.
  - Why don't we use traditional method in ML to cover off-policy?

- To estimate return $J(\pi)$, your IS estimator is

$$\mathrm{IS}(\tau) := \left(\prod_{t=0}^{H-1} \frac{\pi(a_t \mid s_t)}{\pi_D(a_t \mid s_t)} \right) \left(\sum_{t=0}^{H-1} \gamma^t r_t \right).$$

- Exponentially bad…

$$\max_{s,a} \frac{\pi(a \mid s)}{\pi_D(a \mid s)} \leq C_{\mathcal{A}},$$

$$|\widehat{J}_{\mathrm{IS}}(\pi) - J(\pi)| \lesssim V_{\max} \sqrt{\frac{(C_{\mathcal{A}})^H}{n} \log \frac{1}{\delta}},$$

Appendix:
Marginal Importance Sampling (MIS)

- Naïve IS in off-policy estimation (OPE) suffers curse of horizon.

- $d^{\pi}/d^{\mathcal{D}}$ is important in Off-RL analysis → Let's learn it!
  - Like GAN, density ratio works as a 'discriminator'

- Tend to be worse than BRM, theoretically.

Appendix: Other Algorithms

- Pessimism/Conservative Algorithm
  - Using LCB version(PEVI)
  - Conservative Q-Learning, etc…
- Policy constraint
  - IQL, BCQ…
- They have pros and cons compare to BRM.

Bellman Residual Minimization

- In statistical ML, we simply want to minimize a loss…
- Define a new loss, (BRM)

$$\mathcal{L}(Q,\pi) \coloneqq \mathbb{E}_{\mathcal{D}}[(Q - \mathcal{T}^{\pi}Q)^2] \text{ (policy evaluation)}$$
$$\mathcal{L}(Q,\pi) \coloneqq \mathbb{E}_{\mathcal{D}}[(Q - \mathcal{T}Q)^2] \text{ (policy optimization)}$$

- And minimize it!
  - General loss version: $\mathcal{L}(f'; f, \pi) \coloneqq \mathbb{E}_{D}[(f'(s,a) - r - \gamma f(s', \pi))^2]$
  - Great! Are we done? It definitely looks like a SL problem!

Bellman Residual Minimization

So many haters…

Why Should I Trust You, Bellman?
The Bellman Error is a Poor Replacement for Value Error

Scott Fujimoto [1 2] **David Meger** [1] **Doina Precup** [1 3] **Ofir Nachum** [2] **Shixiang Shane Gu** [2]

Bellman error is bad, so BRM is also bad.

Toward Efficient Gradient-Based Value Estimation

Arsalan Sharifnassab [1] **Richard Sutton** [1]

Mean-square BE is ill-conditioned.

11.6 The Bellman Error is Not Learnable

From Sutton's Textbook

BRM: PROS

- No iteration, direct optimization
- No density-ratio estimation
  - Though in many cases we need coverage assumption anyway
- No explicit pessimism penalty / conservatism
- No cancellation (Because it is essentially sum of squared loss)

- Simple!

BRM's biggest CONS: double-sampling

- True Bellman Error =

$$\mathcal{E}(f;\pi) = \mathbb{E}_{(s,a)\sim d^{\mathcal{D}}}\left[\left(f(s,a) - \mathbb{E}_{r,s'|s,a}[r + \gamma f(s',\pi)]\right)^2\right],$$

- Unfortunately, even with infinite dataset it is not a true loss

$$\mathcal{L}(f;f,\pi) = \mathcal{E}(f;\pi) + \mathcal{L}(\mathcal{T}^{\pi} f;f,\pi).$$

  - Simply because $\mathbb{E}[(Y - h(X))^2] = \mathbb{E}[(\mathbb{E}[Y \mid X] - h(X))^2] + \mathbb{E}[(\mathbb{E}[Y \mid X] - Y)^2],$
  - $\mathcal{T}^{\pi} f$ depends on f!
- Moreover, we don't have that! (we usually don't have transition)

Dual Approach and its applications

Setting: entropy-regularized RL

- Add entropy regularizer: $H(q) = -\sum_{a \in \mathcal{A}} q_a \log q_a$

$$\pi^* := \operatorname{argmax}_{\pi \in \Delta_{\mathcal{A}}^{\mathcal{S}}} \mathbb{E}_{\nu_0, \pi} \left[\sum_{h=0}^{\infty} \gamma^h \Big(r(s_h, a_h) + \lambda H(\pi(\cdot \mid s_h)) \Big) \right],$$

$$V^*(s) := \max_{\pi \in \Delta_{\mathcal{A}}^{\mathcal{S}}} \mathbb{E}_{\nu_0, \pi} \left[\sum_{h=0}^{\infty} \gamma^h \Big(r(s_h, a_h) + \lambda H(\pi(\cdot \mid s_h)) \Big) \,\middle|\, s_0 = s \right],$$

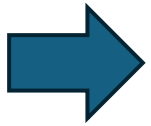
$$Q^*(s,a) := \max_{\pi \in \Delta_{\mathcal{A}}^{\mathcal{S}}} \mathbb{E}_{\nu_0, \pi} \left[r(s_0, a_0) + \sum_{h=1}^{\infty} \gamma^h \Big(r(s_h, a_h) + \lambda H(\pi(\cdot \mid s_h)) \Big) \,\middle|\, s_0 = s, a_0 = a \right].$$

- One can prove that:

$$\pi^*(a \mid s) = \frac{\exp\left(Q^*(s,a)/\lambda\right)}{\sum_{a' \in \mathcal{A}} \exp\left(Q^*\left(s,a'\right)/\lambda\right)}, \quad \forall a \in \mathcal{A},$$

$$V^*(s) = \lambda \ln \left[\sum_{a \in \mathcal{A}} \exp\left(Q^*(s,a)/\lambda\right) \right],$$

$$Q^*(s,a) = r(s,a) + \gamma \mathbb{E}_{s' \sim P(s,a)} \left[\lambda \ln \sum_{a' \in \mathcal{A}} \exp\left(Q^*\left(s',a'\right)/\lambda\right) \,\middle|\, s, a \right].$$



Naturally, for arbitrary Q, define (lambda=1 for convenience)

$$V^Q(s) := \ln \left[\sum_{a \in \mathcal{A}} \exp\left(Q(s,a)\right) \right]$$

$$\pi^Q(a|s) := \frac{\exp\left(Q(s,a)\right)}{\sum_{a \in \mathcal{A}} \exp\left(Q(s,a)\right)}.$$

Back to the Bellman Residual

- Eventually, we want to minimize $\mathcal{L}(Q,\pi) \coloneqq \mathbb{E}_{\mathcal{D}}[(Q-\mathcal{T}Q)^2]$
- Let's think about it individually: $\mathcal{L}_{BE}(Q)(s,a) = (Q-\mathcal{T}Q)^2(s,a)$
  - We don't have $\mathcal{T}Q = r + \gamma \mathbb{E}_{s'}[V^Q(s')]$
  - So, we use $\hat{\mathcal{T}}Q(s,a,s') = r + \gamma\,[V^Q(s')]$.
  - Let $\mathcal{L}_{TD}(Q)(s,a,s') = \left(Q-\hat{\mathcal{T}}Q\right)^2(s,a,s')$

- Problem: Jensen's curse! $\mathbb{E}_X[X^2] \neq \mathbb{E}_X[X]^2$
  - $\mathbb{E}[Q-\mathcal{T}Q] = \mathbb{E}\left[Q-\hat{\mathcal{T}}Q\right]$, but not for squared version!
- How to avoid it?

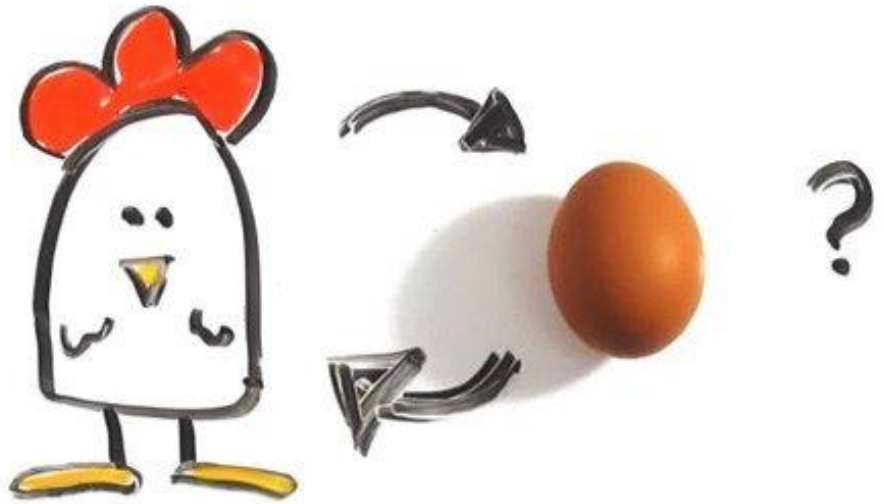
Dual approach

- Surprising fact: $y^2 = min_x\ (2yx - x^2)$ (obvious)

$$\mathcal{L}_{BE}(Q)(s,a) = \mathbb{E}_{s' \sim P(s,a)}\left[\mathcal{L}_{TD}(Q)(s,a,s')\right] - \gamma^2 \min_{\xi \in \mathbb{R}^{S \times A}} D(Q)(s,a,\xi)$$

$$\textit{where } D(Q)(s,a,\xi) := \mathbb{E}_{s' \sim P(s,a)}\left[(V^Q(s') - \xi)^2\right].$$

- Great! Now
  - If we have nice ξ, we can train Q.
  - But, to estimate ξ, we need Q.



GLADIUS algorithm

Operations Research 2025 (Kang et al.),

NeurIPS 2026 (Kang & Jang)

Under Review (Park et al.)

Collaborators



Hyunwook Enoch Kang
Univ. of Washington
(Postdoc)



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Chung-Ang University
(Master)



Minhyeok Park
Chung-Ang University
(Bachelor)

GLADIUS algorithm (OR 2025)

- Originally for IRL and DDC

- Offline-RL version: minimize

$$\frac{1}{N} \sum_{(s,a,s') \in \mathcal{D}_N^b} \left[\mathcal{L}_{\mathrm{TD}}(Q)(s,a,s') - \gamma^2 \left(V^Q(s') - \bar{\xi}(s,a) \right)^2 \right]$$

where

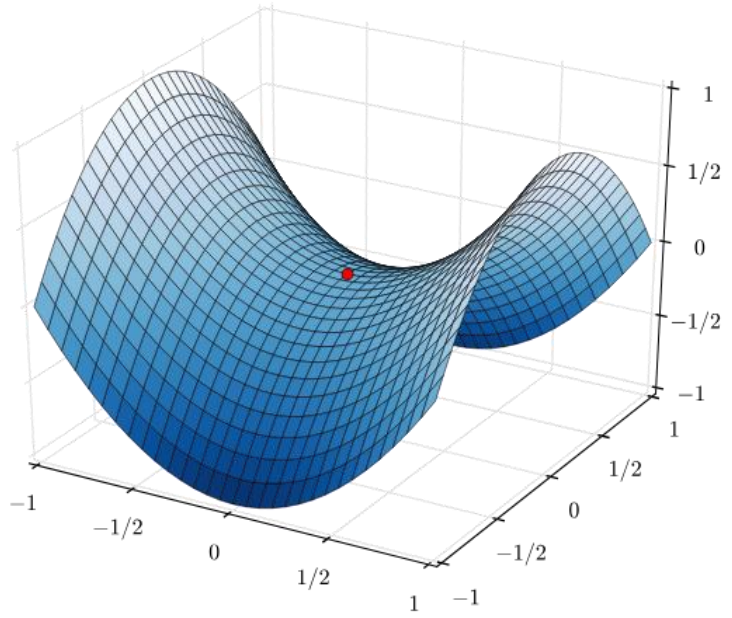
$$\bar{\xi} := \mathrm{argmin}_{\xi \in \mathbb{R}^{S \times A}} \frac{1}{N} \sum_{(s,a,s') \in \mathcal{D}_N^b} \left[\left(V^Q(s') - \xi(s,a) \right)^2 \right]$$

So, how to update your model? SGDA

Algorithm 1 `Off-GLADIUS`

Require: Offline dataset $\mathcal{D} = \{(s, a, r, s')\}$, horizon T, ascent steps K_ξ
Ensure: $\widehat{Q}, \widehat{\xi}$

Initialize $Q_{\theta_1}, \xi_{\theta_2}$; $t \leftarrow 1$
while $t \leq T$ **do**
  Ascent step (update ξ_{θ_2}):
  for $k = 1, \ldots, K_\xi$ **do**
    Sample mini-batch $B_2 \subset \mathcal{D}$
    $D_{\theta_2} \leftarrow \frac{1}{|B_2|} \sum_{(s,a,s') \in B_2} \big(V_{\theta_1}(s') - \xi_{\theta_2}(s,a)\big)^2$
    $\theta_2 \leftarrow \theta_2 - \tau_{2,t} \nabla_{\theta_2} D_{\theta_2}$
  end for
  Descent step (update Q_{θ_1}):
    Sample mini-batch $B_1 \subset \mathcal{D}$
    $\overline{\mathcal{L}_{\mathrm{BE}}} \leftarrow \frac{1}{|B_1|} \sum_{(s,a,s') \in B_1} \Big[\mathcal{L}_{\mathrm{TD}}(Q_{\theta_1})(s,a,s') - \gamma^2 \big(V_{\theta_1}(s') - \xi_{\theta_2}(s,a)\big)^2\Big]$
    $\theta_1 \leftarrow \theta_1 - \tau_{1,t} \nabla_{\theta_1} \overline{\mathcal{L}_{\mathrm{BE}}}$
  $t \leftarrow t + 1$
end while
Return $\widehat{\xi} \leftarrow \xi_{\theta_2}$, $\widehat{Q} \leftarrow Q_{\theta_1}$



Previous work: SBEED (Dai et al., 2019)

- State value function (V) version of this duality

$$\min_{V} \mathbb{E}_{s\sim\mu}\left[\left(\max_{\pi(\cdot|s)\in\mathcal{P}_{\mathcal{A}}} \mathbb{E}_{a\sim\pi(\cdot|s)}\left[R(s,a)+\gamma\mathbb{E}_{s'|s,a}\left[V(s')\right]\right]-V(s)\right)^2\right]$$

- Train three parameters: V, π, and ρ (essentially ξ of ours)
  - On the other hand, when Q is fixed, V and π are fixed functions.

Improvement

- Previously, there were only limited convergence results:

$$\mathbb{E}\left[\left\|\nabla\widehat{\ell}_T(\widehat{V}_w^N,\widehat{\pi}_w^N)\right\|^2\right] \leqslant \frac{KD^2}{N} + (D' + \frac{D}{D'})\frac{\sigma}{\sqrt{N}}$$

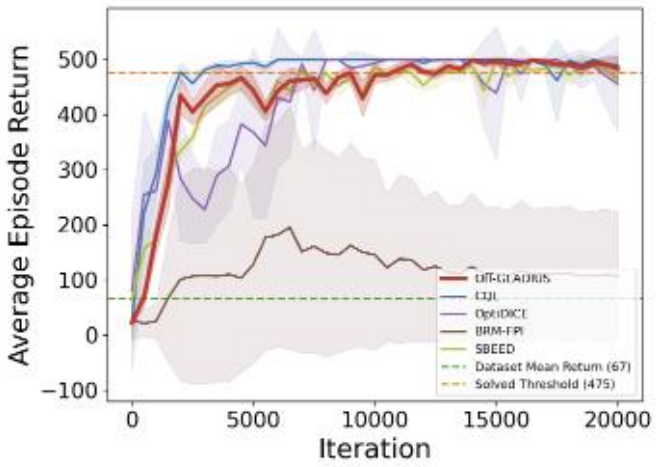
  - (N iterations)

- (Off)-GLADIUS achieves, under relatively general conditions (large neural network, linear), guarantees **global convergence**!
  - (T iterations, N samples)

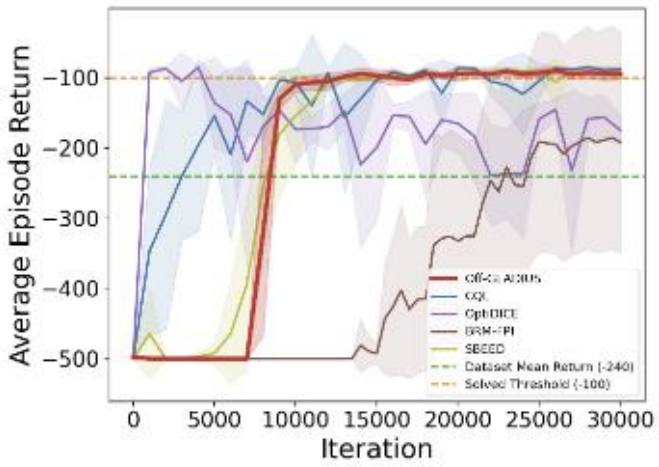
$$\mathbb{E}\left[\|Q_{\hat{\theta}_T} - Q^*\|_{2,\pi^*}^2\right] = O\big((c_2+T)^{-\alpha}\big) + O\left(\frac{1}{N}\right) + O\left(\frac{1}{T}\right)$$

Experiment: better results

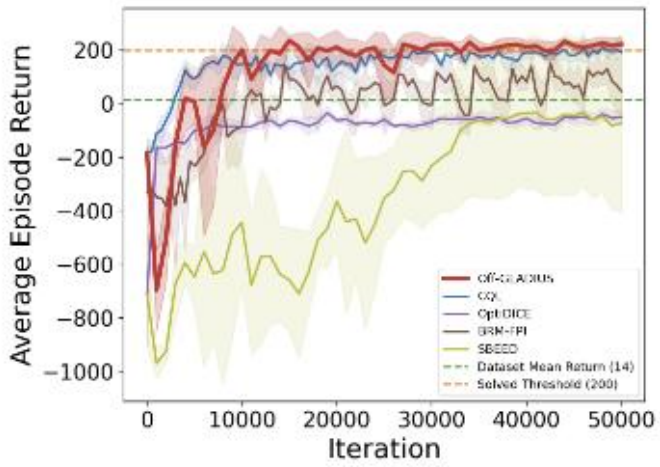
Our algorithm **(red)** converges stably compared to other algorithms.



(a) Cartpole-v1



(b) Acrobot-v1



(c) LunarLander-v3

Statistical loss? (NeurIPS 2026)

- One can decompose loss by ϵ_{opt}& ϵ_{stat}
  - Optimization error (ϵ_{opt}): Caused by limited (finite) optimization steps
  - Statistical error (ϵ_{stat}): Caused by limited (finite) samples

$$R(Q_T) - R(Q^\star) = \underbrace{R(Q_T) - R_N(Q_T)}_{\text{Statistical / generalization}} + \underbrace{R_N(Q_T) - R_N(\widehat{Q})}_{\text{Optimization}} + \underbrace{R_N(\widehat{Q}) - R(Q^\star)}_{\text{Remaining comparison}}.$$

- Though Off RL can be understood as Stat. ML, statistical loss results were not good.

Previously…

- Worse rates

Reference	Setting	Rate
Malliard et al., 2008	BRM (Linear V)	$O(n^{-1/2})$
Wang et al., 2022	Convex-concave	$O(n^{-1/2})$
Dai et al., 2018	BRM (V, π)	$O(n^{-1/2})$
Ours	BRM (Q)	$O(n^{-1})$

Previously…

- Or good rates under the limited settings

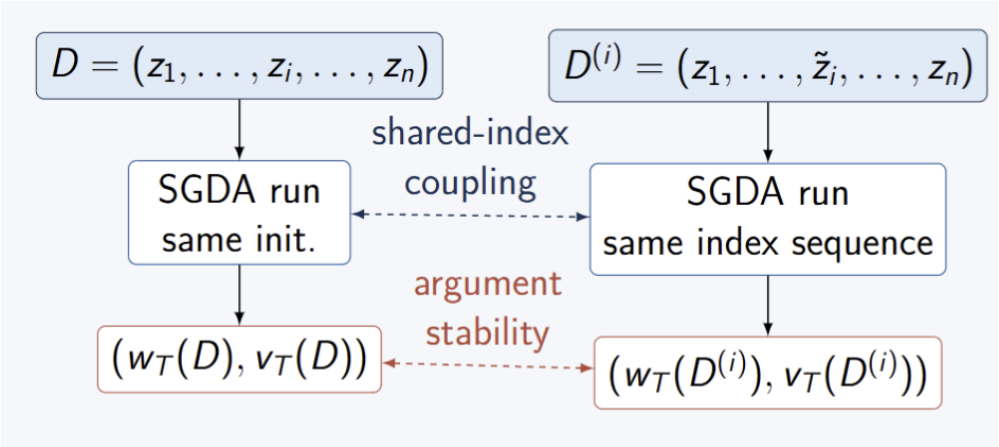
References	Setting	Assumption	Rate
Chen and Jiang (2019)	BRM (MSBO), Q: finite function class	Realizability Coverage Completeness	$O(\frac{1}{n})$
Hu et al (2023)	BRM (MSBO), Q: linear function class	Realizability Coverage Completeness	$O(\frac{1}{n})$
Dual et al (2021)	BRM Q: RKHS	Realizability Coverage ‘for all intermediate steps’ Completeness	$O(\frac{1}{\sqrt{n}})$

Our work (NeurIPS 2026)

- 1/n stat. error rate, even for relatively general cases (wide NN)
  - At the cost of: Optimization error loss

Corollary 4 (Informal bound under a harmonic stepsize schedule). *Choose the Robbins–Monro stepsizes in* ALGORITHM 1 *as* $\eta_t = \frac{c_1}{c_2+t}, t \geq 0$, *with constants* $c_1 > 0$ *and* $c_2 \geq 1$ *small enough that satistying (S1). Let* $c := \min\{\mu_{\mathrm{PL}}/2, \rho/2\}$. *Then from Theorem 3, the stability constant* ε_T *in Theorem 3 admits the stability bound scales as*

$$\varepsilon_T = O\left((c_2+T)^{-\min\left\{\frac{1}{2}, \frac{3c\,c_1}{8}\right\}}\right) + O\left(\tfrac{1}{n}\right).$$



Trajectory-BRM

Feedbacks are expensive...

- In real dataset, we only observe a reward for each episode.
- On the other hand, Offline RL requires stepwise loss.



Trajectory BRM (Jia et al., 2025)

- In LLM, after an action, next state is **deterministic!**
  - Ex) "I am" + a → "I am a"

- Therefore, no double-sampling problem in LLM models.

$$\mathcal{E}(f;\pi) = \mathbb{E}_{(s,a)\sim d^D}\left[\left(f(s,a) - \mathbb{E}_{r,s'|s,a}[r + \gamma f(s',\pi)]\right)^2\right],$$

- Can enjoy advantages of BRM without problems.

Trajectory BRM (Jia et al., 2025)

- Classic BRM (Step-level): minimizes error at every token

$$\mathcal{L}_{\mathcal{D}}^{\mathsf{BRM}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \sum_{h=1}^{H} \Big(\mathsf{logit}_\theta(s_h, a_h) - (\mathcal{T}_\beta \mathsf{logit}_\theta)(s_h, a_h) \Big)^2$$

$$= \frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \sum_{h=1}^{H} \left(\mathsf{logit}_\theta(s_h, a_h) - \frac{r(s_h, a_h)}{\beta} - \log \pi_{\mathsf{ref}}(s_h, a_h) - \log \sum_{a \in \mathcal{A}} \exp(\mathsf{logit}_\theta(s_{h+1}, a)) \right)^2$$

Fails in LLMs due to sparse outcome rewards.

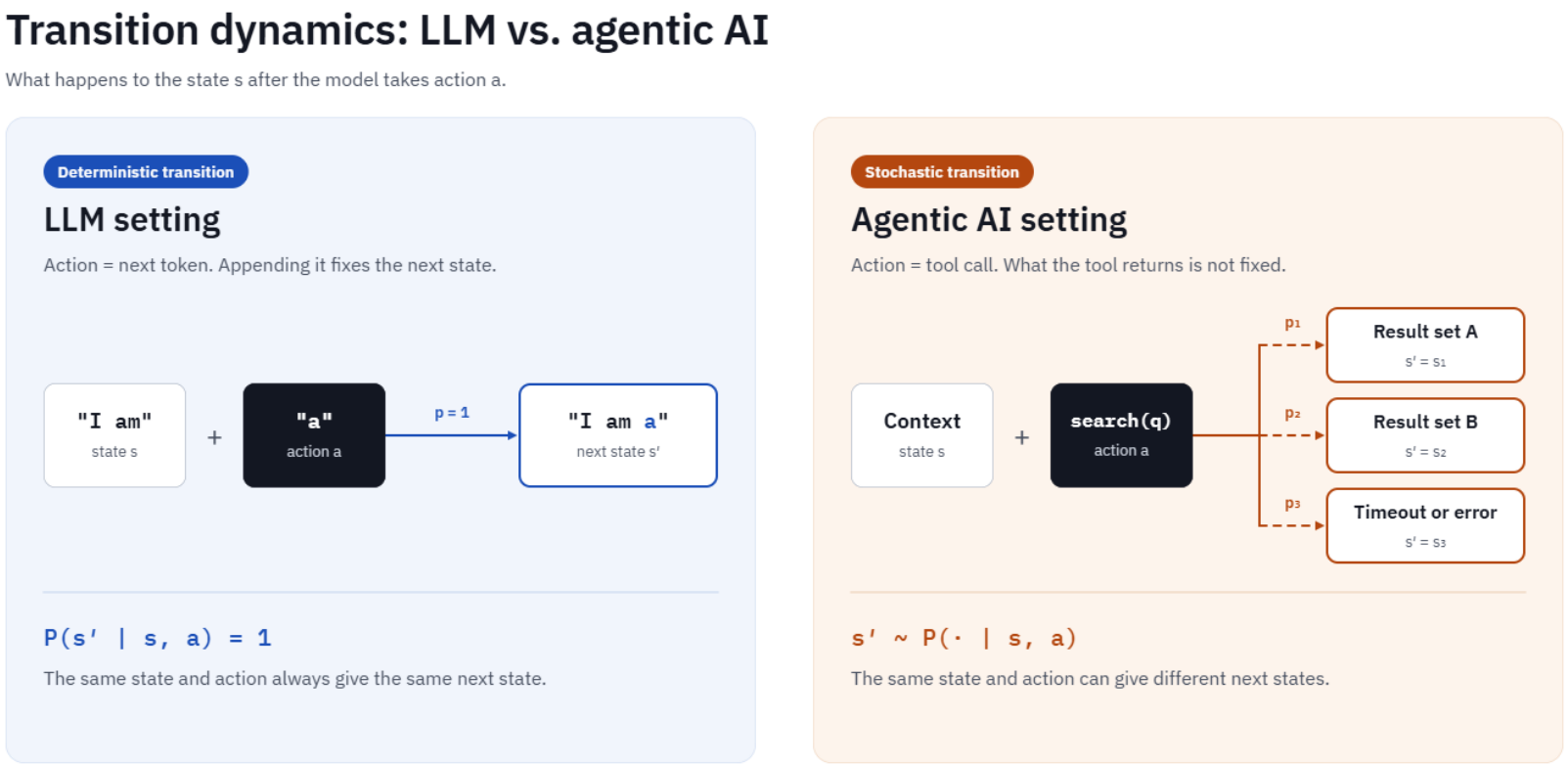
- Trajectory BRM: Lifts the residual to the entire trajectory.

$$\mathcal{L}_{\mathcal{D}}^{\mathsf{TBRM}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \left(\sum_{h=1}^{H} \mathsf{logit}_\theta(s_h, a_h) - (\mathcal{T}_\beta \mathsf{logit}_\theta)(s_h, a_h) \right)^2$$

$$= \frac{1}{|\mathcal{D}|} \sum_{\tau \in \mathcal{D}} \left(\mathsf{logit}_\theta(s_1, a_1) - \frac{r(\tau)}{\beta} - \log \pi_{\mathsf{ref}}(\tau) + \sum_{h=2}^{H} \log \pi_\theta(a_h \mid s_h) \right)^2$$

In agentic AI setting…

- Results from tools are usually **stochastic!**



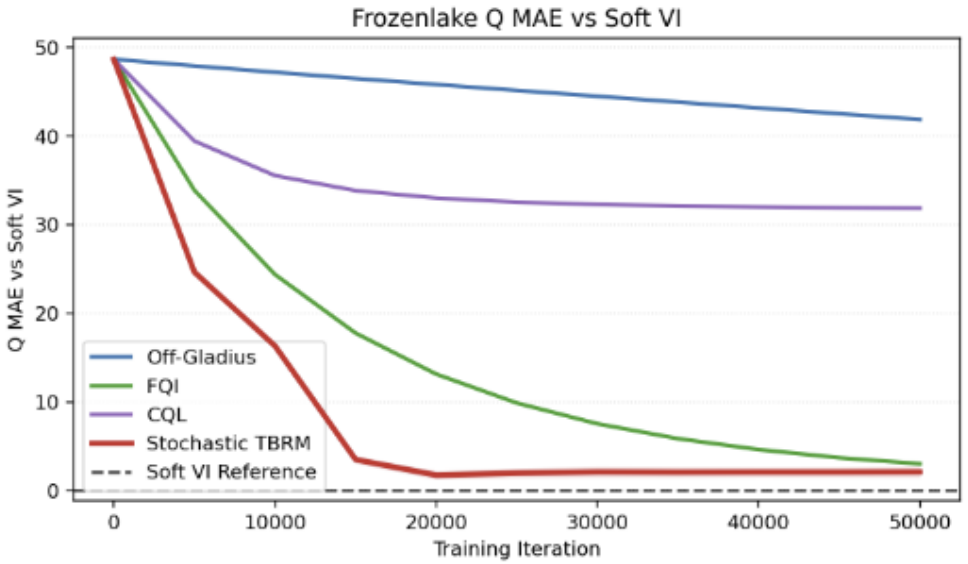
Facing double sampling problem

- **Solution:** dual approach, again!

$$\widehat{R}_{Q,\xi}(s,a) := Q(s,a) - \gamma\xi(s,a), \qquad G_{Q,\xi}(\tau) := \sum_{h=0}^{H-1} \gamma^h \widehat{R}_{Q,\xi}(s_h,a_h).$$

$$\min_{Q\in\mathcal{Q}} \mathbb{E}_{\tau\sim\mathbb{P}^{\pi_b}}\left[\left(G_{Q,\xi^Q}(\tau) - R(\tau)\right)^2\right], \quad \text{where} \quad \xi^Q \in \arg\min_{\xi} \mathcal{J}(Q,\xi).$$

Results



Benchmark	Method	Episodes	Reward	Completion	Steps
WebShop	DMPO	200	0.649	0.955	**3.97**
	Q-Evolve	200	0.674	0.940	4.51
	Stochastic-TBRM (Ours)	200	**0.682**	**1.000**	4.03
ScienceWorld	DMPO	211	0.405	0.156	**19.56**
	Q-Evolve	211	0.520	0.265	21.95
	Stochastic-TBRM (Ours)	211	**0.539**	**0.308**	26.21
ALFWorld	DMPO	134	0.828	0.828	19.07
	Q-Evolve	134	0.463	0.463	25.71
	Stochastic-TBRM (Ours)	134	**1.000**	**1.000**	**10.01**

$$J(\pi^\star) - J(\pi_{\widehat{Q}}) \leq \widetilde{O}\left(\sqrt{C_{\mathrm{conc}}(\pi_b)\left(\varepsilon_{\mathrm{opt}} + \frac{\dim(\Theta)}{n}\right)}\right).$$

Conclusion

- BRM is a SL-like approach of Offline RL.
  - Main advantage: Free from iterativeness
  - Main problem of BRM: double sampling

- Duality can solve double-sampling issue!
  - GLADIUS algorithm

- BRM is still good!
  - Many other low-hanging fruits…

Thank you!

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